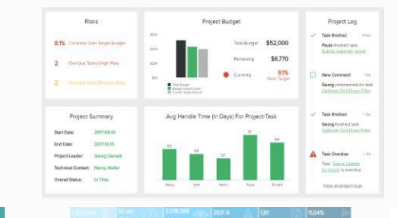
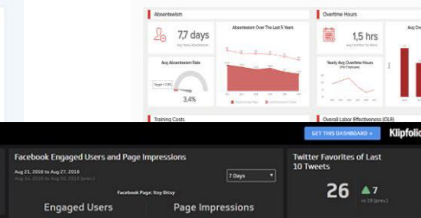
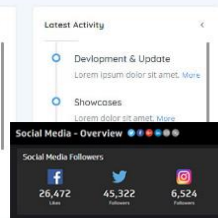
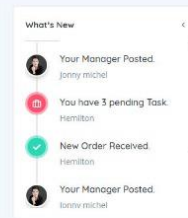
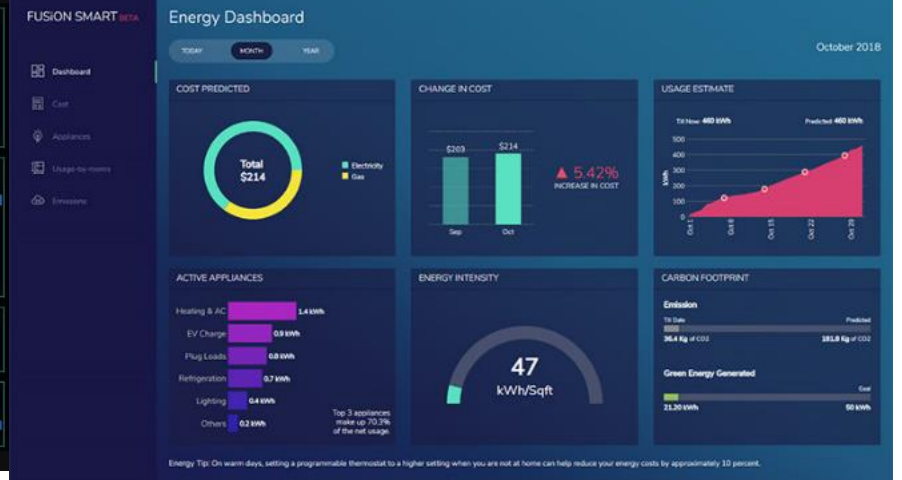
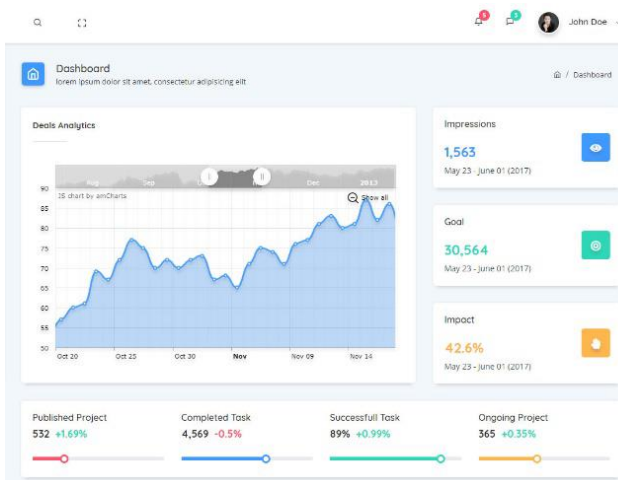


Human Factors and Human-Machine Interaction Information Dashboards

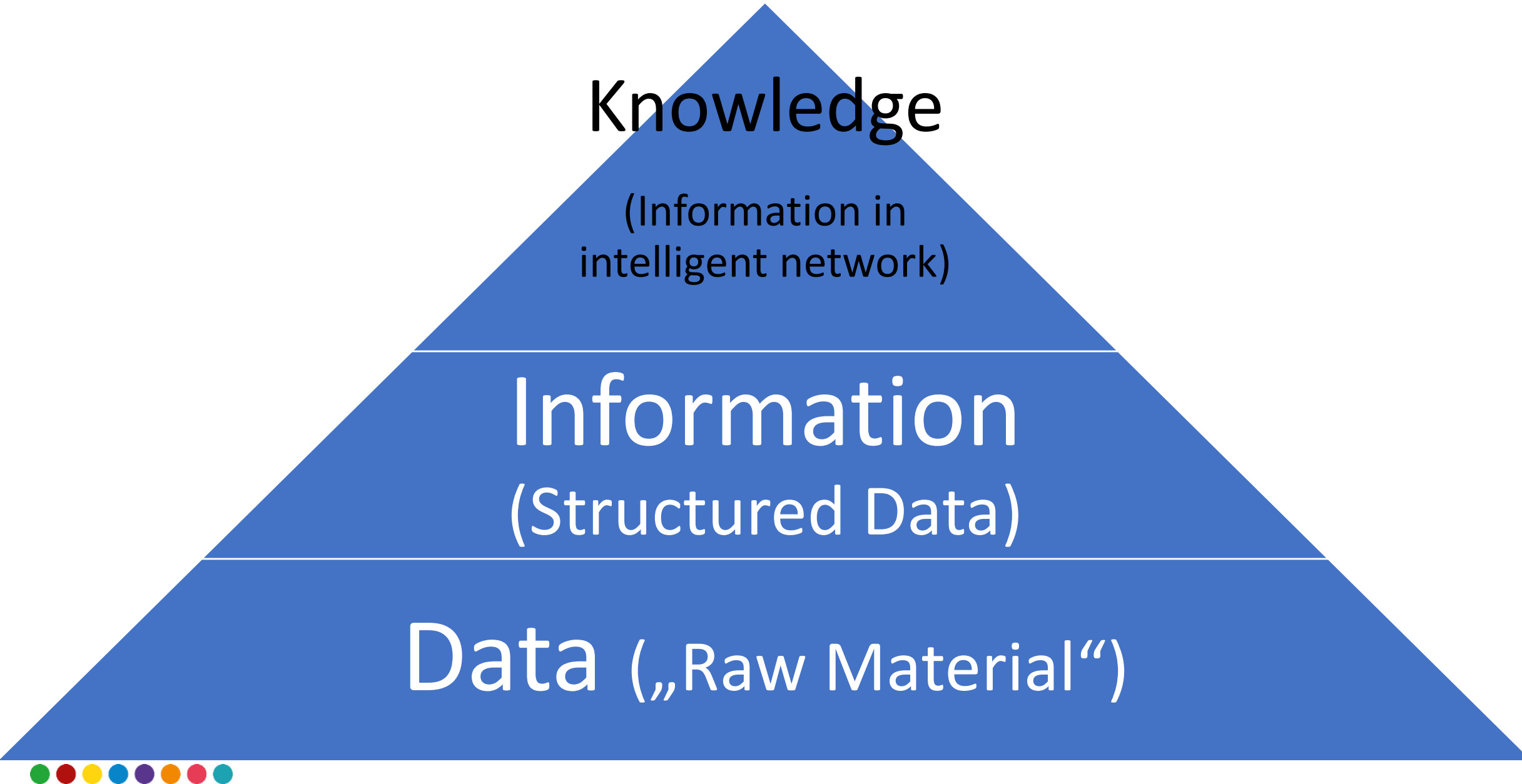
FACULTY
OF COMPUTER SCIENCE





Why should computer science become cognitive?





Knowledge

(Information in
intelligent network)

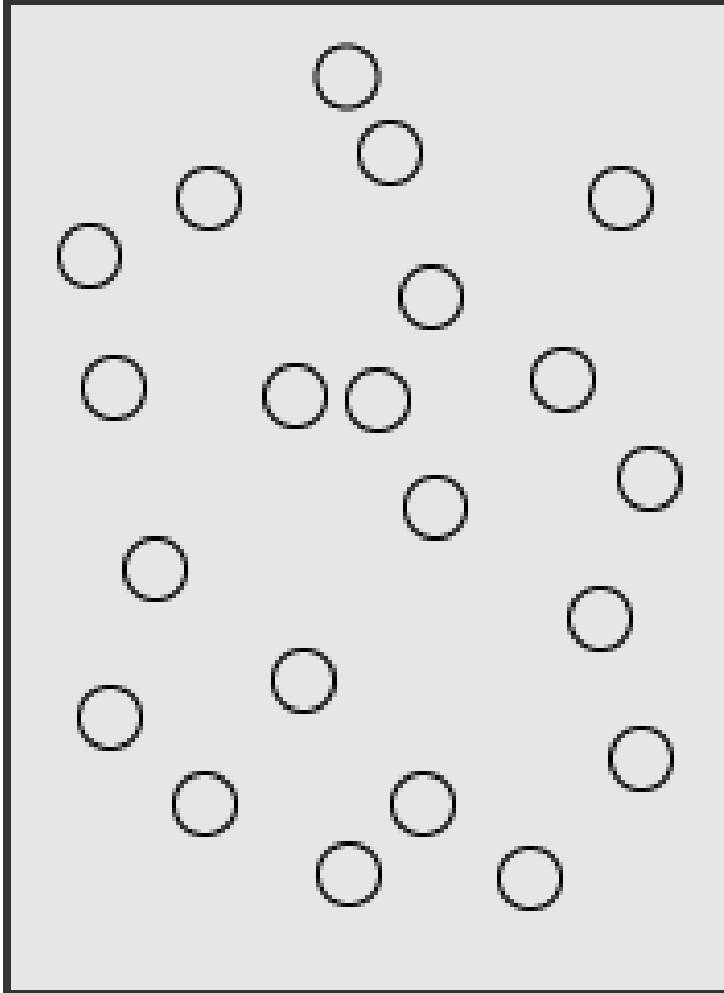
Information

(Structured Data)

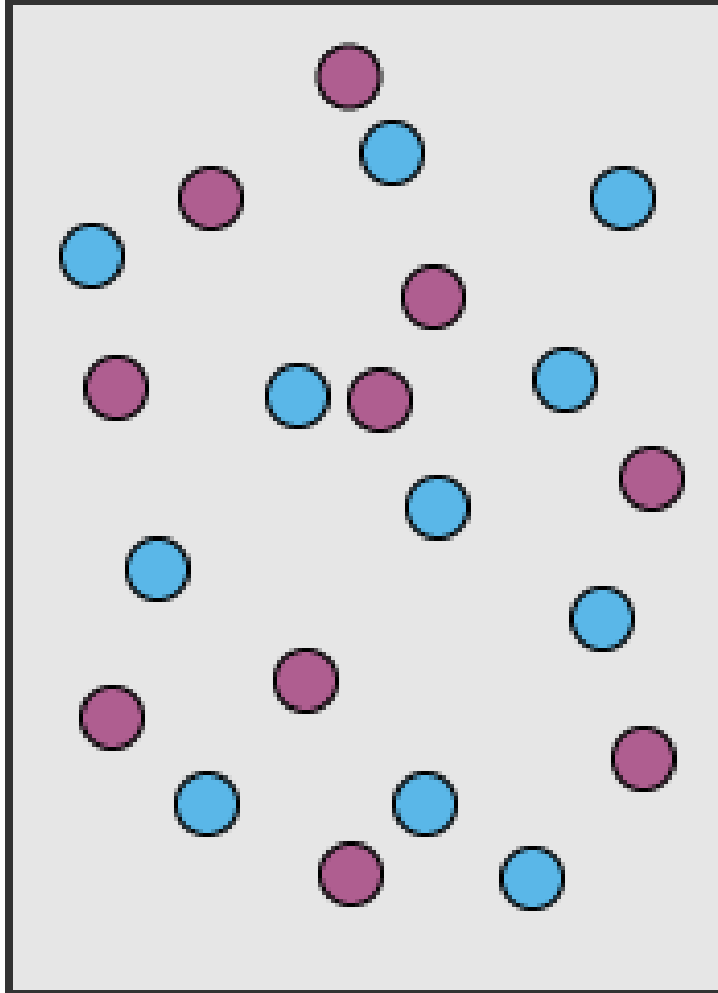
Data („Raw Material“)



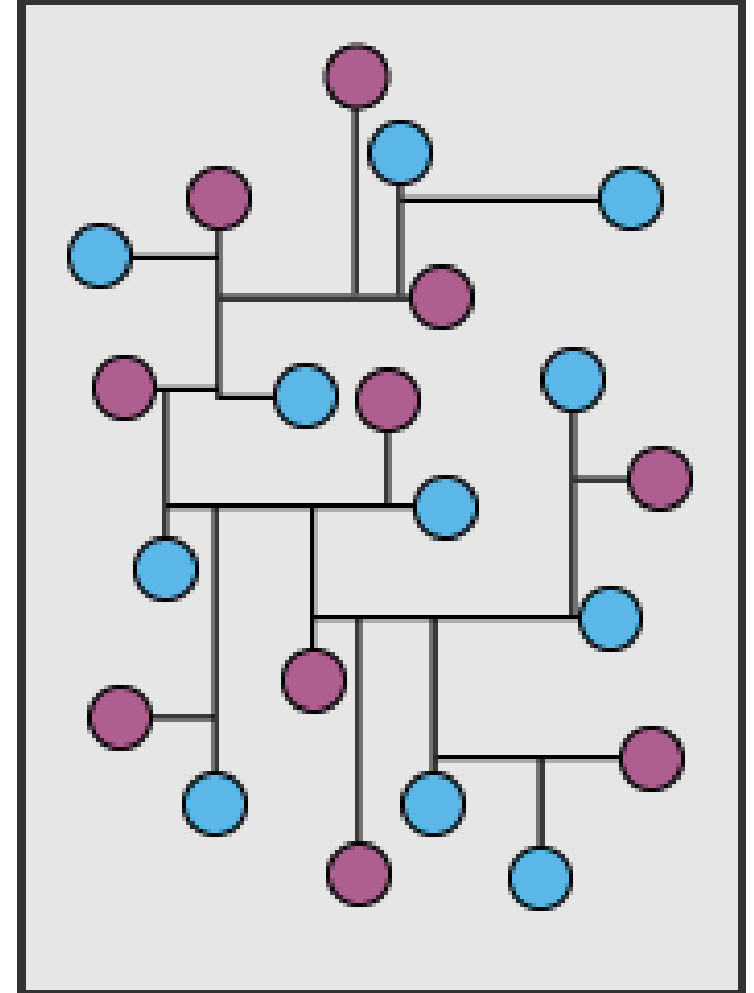
DATA

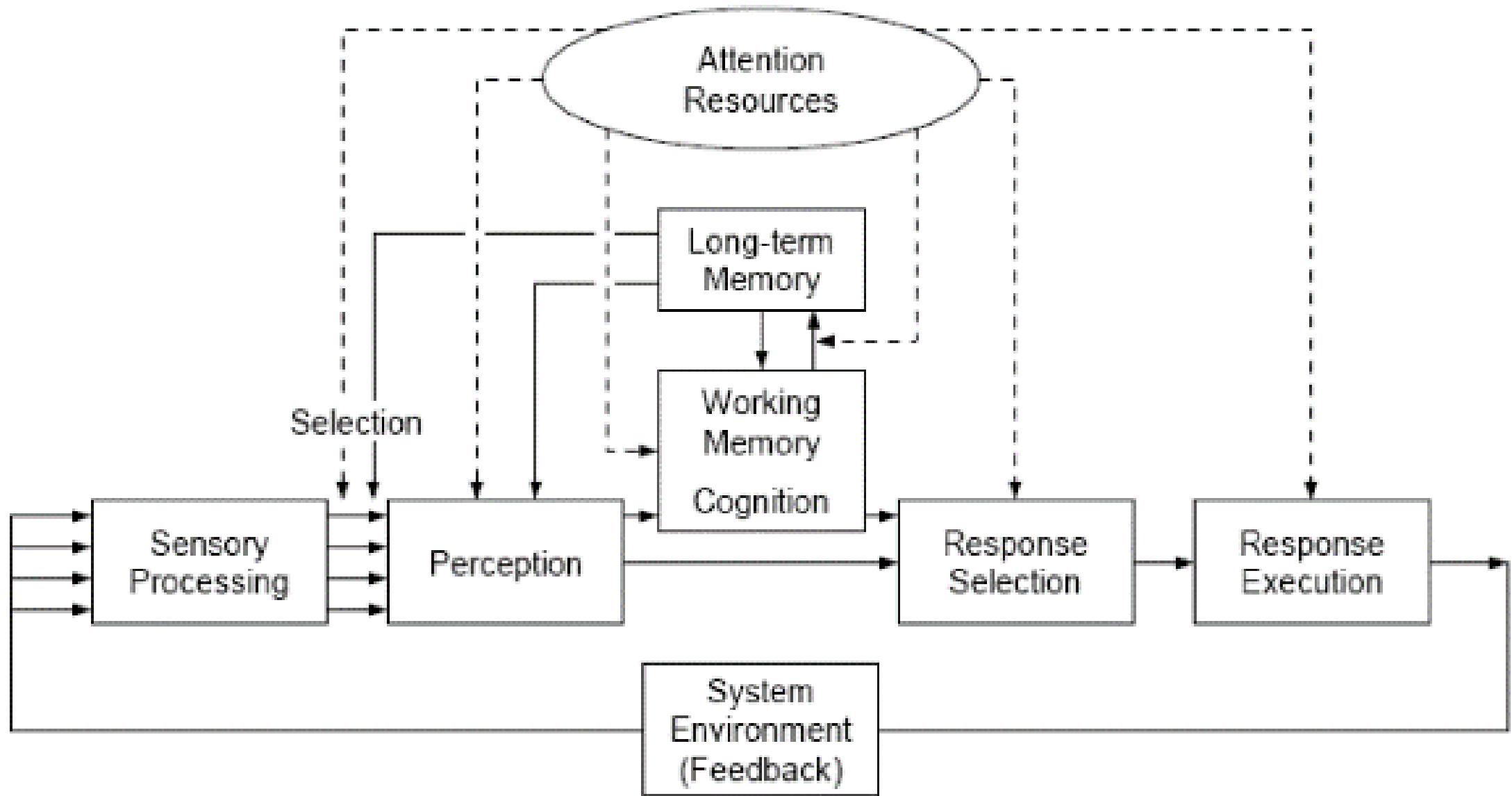


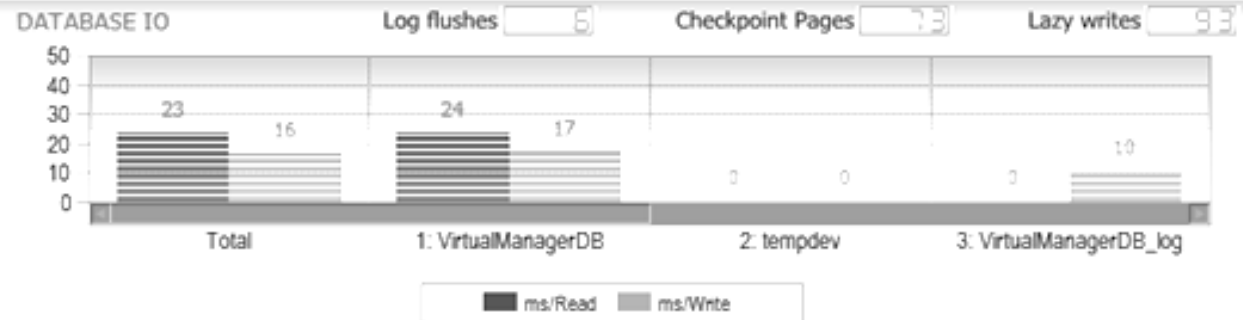
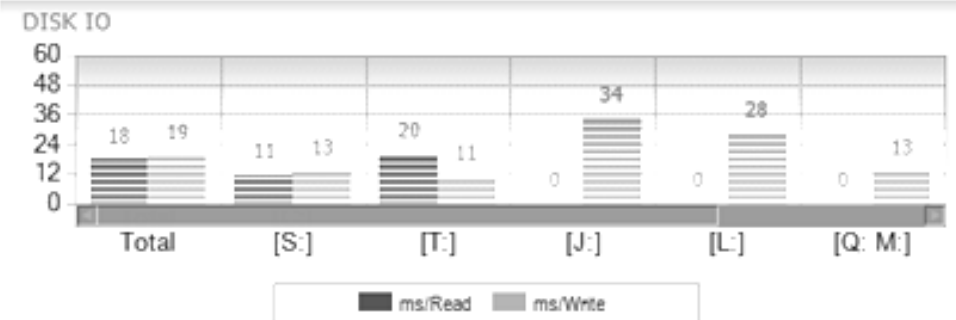
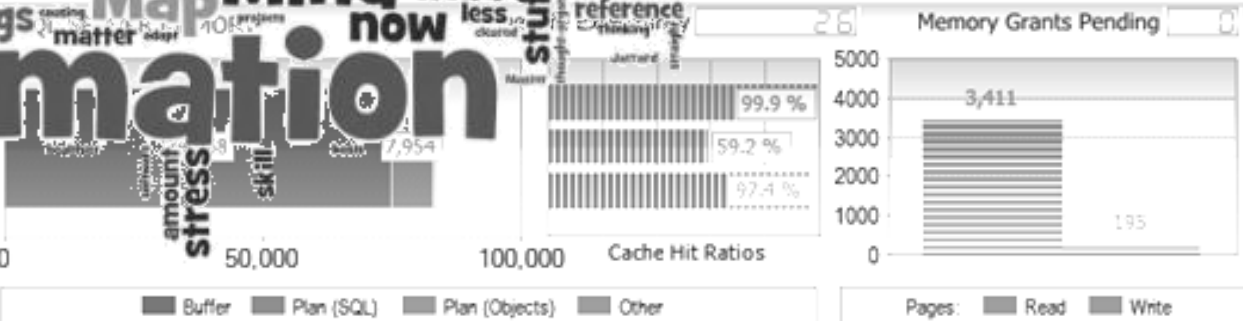
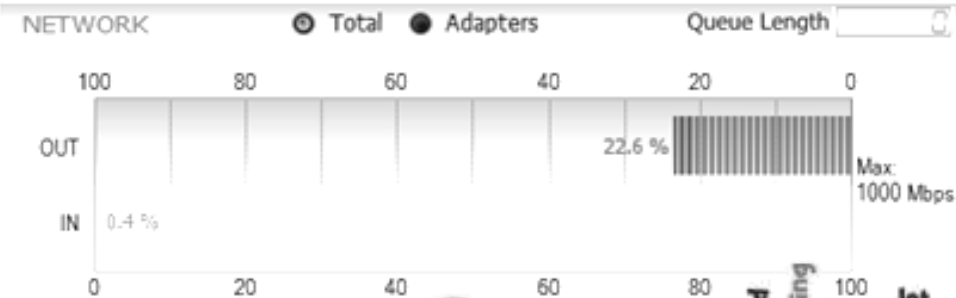
INFORMATION



KNOWLEDGE







information
Overload
Mind
Information
people
day
time
new
project
point
running
reference
stuff
now
less
stress
amount
Map
organise
business
meeting
need
recall
productive
problem
needs
just
detailed
retain
plan
want
may
challenge
overload
make
understand
skills
apply
absorb
done
everyone
busy
important
capture
control
start
read
see
lot
learn
ending
find
small
never
life
much
successful
use
Goddard
easy
many
planning
focus
develop
means
pretty
using
Paper
Targets
factor
power
Mindware
burst
hard
loss
simple
work
things
matter
adap
100%

HERBERT A. SIMON



„The design of systems must encompass far more than hardware and software; it must consider with equal care the information processing capabilities of the human members of organizations who form the other half of the system.“ (1968)



HERBERT A. SIMON



„In the post-industrial society, the central problem is (...) how to organize information to make decisions.“

Simon, H.A. (1973). *Administrative Behavior. A Study of Decision Making Processes in Administrative Organization*, New York, London.



- Meldung
- Ticketplanung
- Einsatz
- Ticketübersicht
- Meine Aufgaben
- Plantafel
- Multiprojektplan-Erstellung
- Projektbewertung
- Projekt-Rechnungslauf
- Ressourcenplan-Erstellung
- Archiv
- Berichte
- Einrichten

- Mein Menü
- Management & Controlling
- Einkauf
- Lager & Logistik
- Verkauf & Marketing
- Service & Werkstatt
- Projektmanagement
- Personalwesen

01BG-0025	Sales Media Research	Calls (2.00 Calls Juni)
01BG-0025	Sales Media Research	Calls (3.00 Calls Juli)
01BG-0025	Sales Media Research	Entwicklungen (comp. trees)
P00105	Bau Analyse München DE	Recherche
P00105	Bau Analyse München DE	Recherche
P00105	Bau Analyse München DE	FPDS 01. Hinzunehm

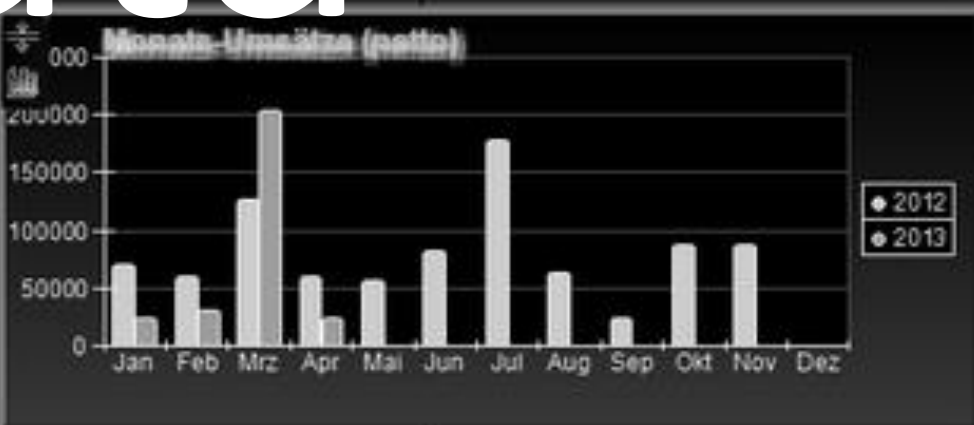
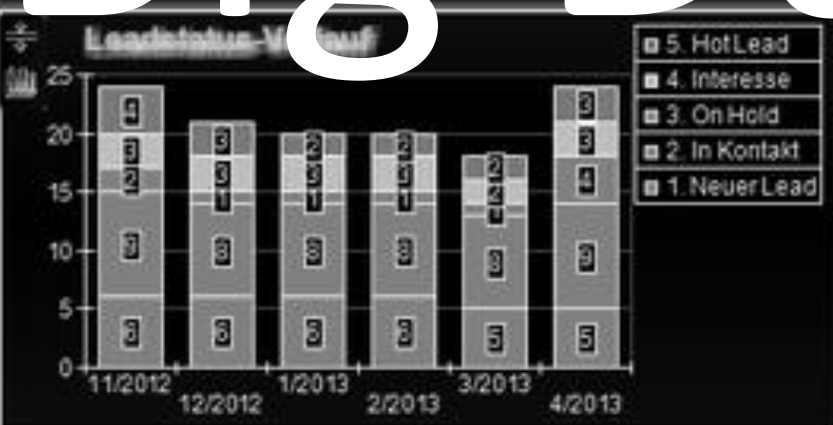
P00109	Twinning	3200.0	32.0	3168.0
P00113	Sicherheitsberatu...	4.0	4.0	0.0
01BG-0025	Sales Media Res...	6502.0	128.4	6373.56
P00105	Bau Analyse Mün...	69.0	12.0	57.0
P00114	Analyse	98.0	0.0	98.0
P00115	Analyse	98.0	0.0	98.0

Projektmanagement: Projektergebnisse

Projektnummer	Projekt	Umsatz	Kosten
P00114	Analyse	780.00	500.0
P00105	Bau Analyse Münch...	1.200.00	20.566.0
P00127	Photovoltaikanlage ...	2.400.00	0.0
P00130	Innenausbau Hotel ...	9.300.00	400.0
P00123	Umbau ...	18.000.00	7.433.0
P00124	Hydraulikregat	19.500.00	1000.0
P00128	...verke...	23.600.00	0.0

Projektmanagement: Budgetstunden-Überwachung

Projektnummer	Projekt	Verbleibend	Budget	Ist
P00113	Sicherheitsberatung	1.0	1.0	0.0
P00105	Bau Analyse München DE	28.0	40.0	12.0
01BG-0025	Sales Media Research	2871.56	3000.0	128.4
P00109	Twinning	3168.0	3200.0	32.0
P00111	Sicherheitsberatung	-3.0	1.0	4.0
P00125	Programmierung Schnittstelle xy	2.3	8.0	5.57
P00123	Umbau ...	-12.0	40.0	52.0



Aktuelle OPOS-Bestände:

Forderungen:	3.806.345,07 €
Verbindlichkeiten:	2.882.179,43 €



Monatsumsatz:

Anzahl Re / Gt:	5 / 0
Volumen netto:	23.720,00 €
Rohertrag:	12.341,00 €



Aktuelle OPOS-Bestände:

Forderungen:	3.806.345,07 €
Verbindlichkeiten:	2.882.179,43 €



Fakturierung:

Anzahl Re / Gt:	5 / 0
Volumen netto:	23.720,00 €
Rohertrag:	12.341,00 €

STATUS QUO

- More and **more data**
- Bottleneck: human information **processing**
- Increasing **challenges for the design** of information-dense displays
- Design must know, take into account and exploit **principles** and **mechanisms** of perception and information processing.
- Daniel Kahneman: „Whatever else it produces, an organization is a factory that manufactures judgments and decisions“ (p. 418)



STATUS QUO

- History & Keywords:
 - Executive Information Systems
 - Decision Support Systems
 - Data Warehousing
 - OLAP
 - Business Intelligence, Business Analytics
 - Key Performance Indicators
- Flashy Dashboards



DASHBOARD DEFINITION

*A dashboard is a **visual** display of the **most important information** needed to achieve one or more **objectives**; consolidated and arranged on a **single screen** so the information can be monitored at a glance. (Few, 2013; p. 26)*



GENERAL REMARKS

- Dashboards **≠** display for **data analysis**/exploration, **scorecard**, report to look up **facts**
- Use **skeuomorphism** purposefully - rather avoid/don't overuse metaphors
- Dashboards are **customized** solutions; *not one-size-fits-all*
- Dashboards support formation and maintenance of **situational awareness**
 - Perception
 - Understanding
 - Anticipation



USER STORIES

User stories are short, simple descriptions of a feature told from the perspective of the user or customer.

- Focus on user needs
- Encourage team communication

Basic characteristics of good user stories (INVEST):

- Independent
- Negotiable
- Valuable
- Estimable
- Small
- Testable



Structure of a User Story:

As a [role]

I want to [goal/feature]

so that [reason/value].

Example: As an online customer, I want to track my order so that I know when it will arrive.

EXERCISE: SKETCH A DIT STUDENT DASHBOARD

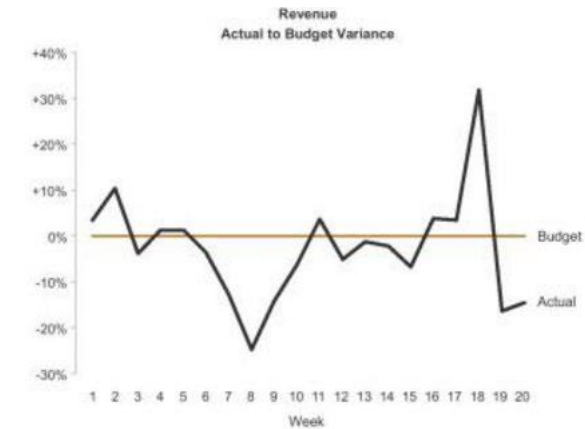
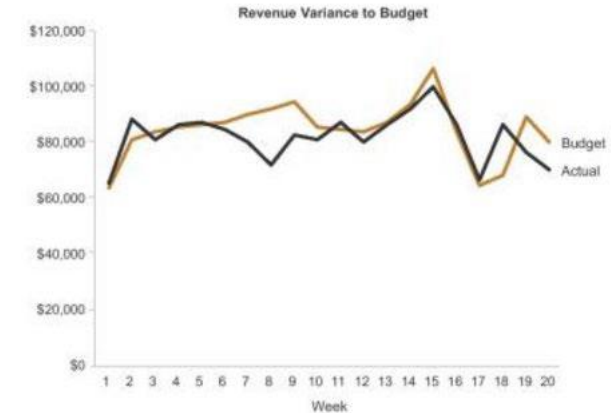
Part 1: Requirements Engineering and Design

1. What information should be displayed on a dashboard? **Write user stories** (at least 5).
2. Create an **initial sketch** of your dashboard having the following questions in mind:
 - Which questions/information have the highest priority? How can these priorities be weighted visually?
 - Is attention appropriately directed (SEEV model)? Which contrasts can be used appropriately?
 - How can you use which gestalt laws appropriately?
 - Are there certain states, events, particularities to which attention must be drawn as soon as they occur?



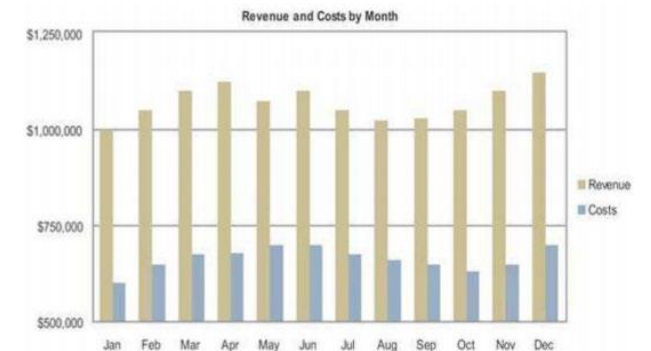
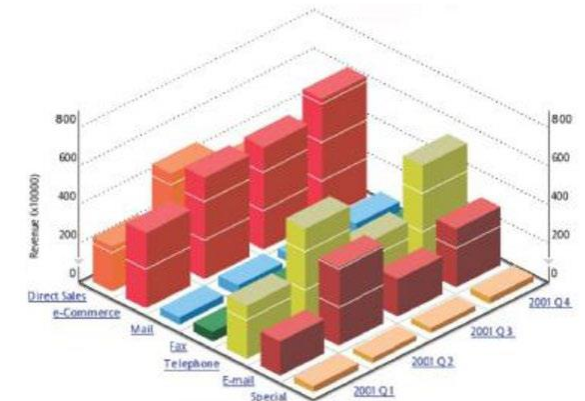
COMMON MISTAKES

- Exceeding the boundaries of a **single screen**
 - Scrolling/fragmented screens: Information Access Costs
 - No overview, no relationships
- Supplying inadequate **context** for the data
 - Comparison biases (baseline, min/max, mean, SD, ...)
 - Measures
 - Evaluations
- Displaying excessive **detail** or precision
 - Suitable rounding (3,301,654.93 EUR vs. 3.3 million EUR)
 - Avoid clutters
- Expressing **measures** indirectly (choosing a deficient measure)
 - Task orientation
 - Absolute values or relative developments?
 - Deviation from reference measure?



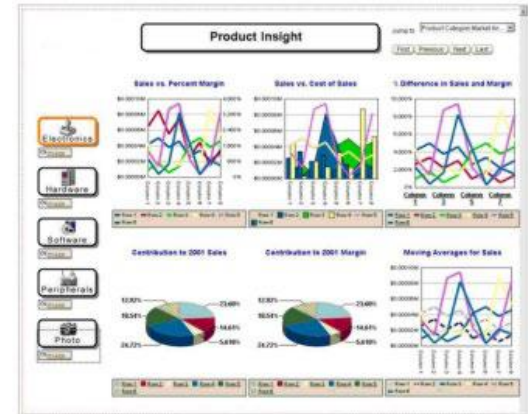
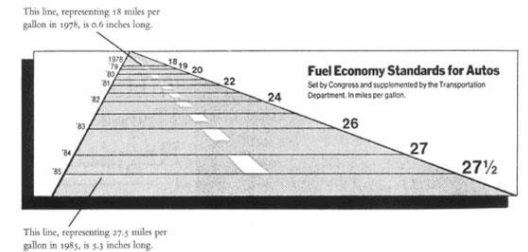
COMMON MISTAKES

- Choosing inappropriate **display media**
 - Diagram instead of table or vice versa
 - Circle diagrams
 - Areas, volumes for exact quantities
- Using poorly designed **display media**
 - Unnecessary legends with colour areas that are too small
 - Structures in the data are not used (e.g., sorting by size)
 - Contrasts are too strong
 - Chart junk, 3-D, ...
- Introducing meaningless **variety**
 - Visual differences should reflect differences in content
 - No different diagrams for similar sizes just for the sake of variety



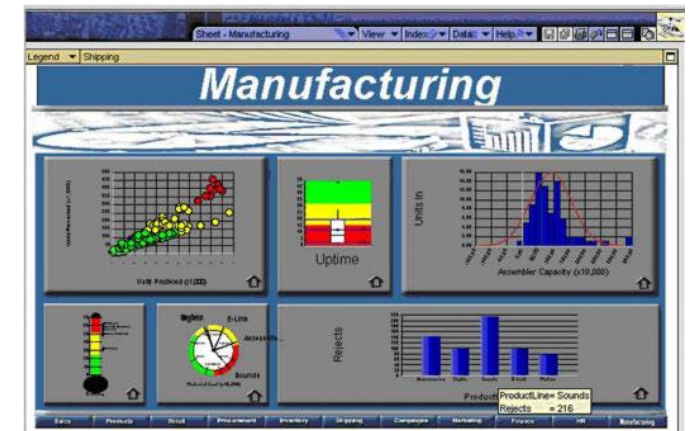
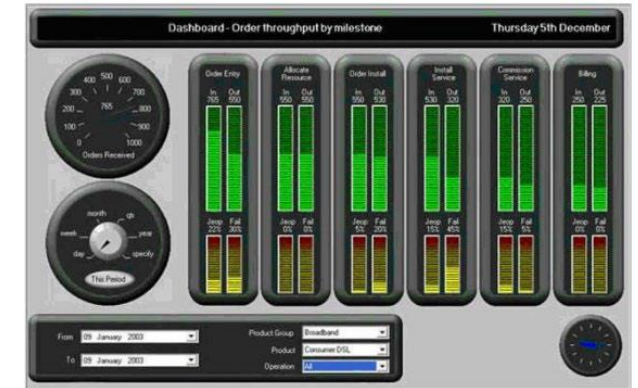
COMMON MISTAKES

- Encoding **quantitative data** inaccurately
 - Visual Lies → Lie Factor = depicted effect size/actual effect size
- **Arranging** the data poorly
 - Ignoring the direction of reading
 - Wasting prominent placements
- **Highlighting** important data ineffectively or not at all
 - Drawing attention to wrong content
 - Too much contrast; if everything is prominently designed, nothing stands out.



COMMON MISTAKES

- **Cluttering** the display with visual effects
 - Decoration
 - Metaphors, skeuomorphism
- **Misusing** or overusing **color**
 - Economical use
 - Generation of references: (un)intentional
 - Use for attention control
- Designing an **unattractive** visual display
 - Aesthetics are not created through decoration
 - Reduced colour saturation
 - White space important
 - C.R.A.P.



REQUIREMENTS ANALYSIS

- Focus on the **goals** not the means/methods
- Identify **mental models** of users:
 - Structures, processes, dependencies
 - Mapping via simple sketches
- Identify **requirements** through questions
 - How often is information/dashboard used?
 - Who is/are the user(s) (expert, novice, layperson)?
 - What objectives does the dashboard need to support?
 - What questions should the dashboard answer?
 - What information in what level of detail; what data types?
 - Which information is important for the goals and how?
 - Which logical groupings exist?
 - What comparisons, benchmarks, assessments are useful and necessary?
 - ...



DATA-INK RATIO

- Data-ink (or data-**pixel**) = "*the ink changing as the data change*" (Tufte, 1983; p. 93)
- Representation of quantitative information should consist mainly of data (data-ink) and as little non-data as possible.
- **Design principle:** *Maximize the data-inkratio, within reason. Every bit of ink on a graphic requires a reason. And nearly always that reason should be that the ink presents new information.* (p. 96)
- **In short:** reduction of unnecessary (non-data) pixels, reduction of unnecessary variability (cf. right o. linear colour gradients), improvement of data pixels.

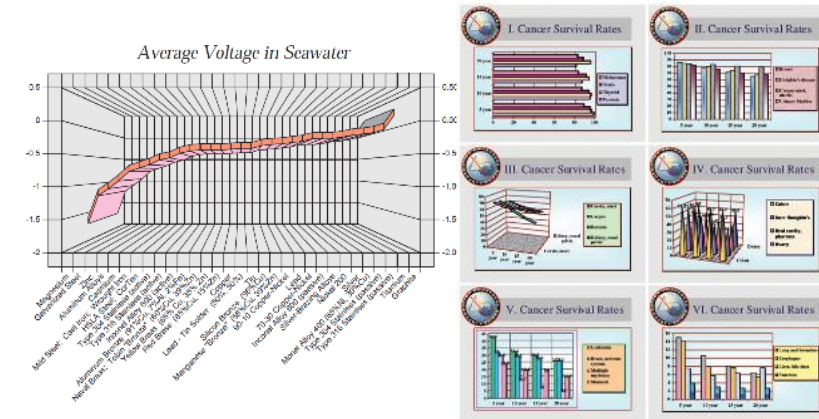
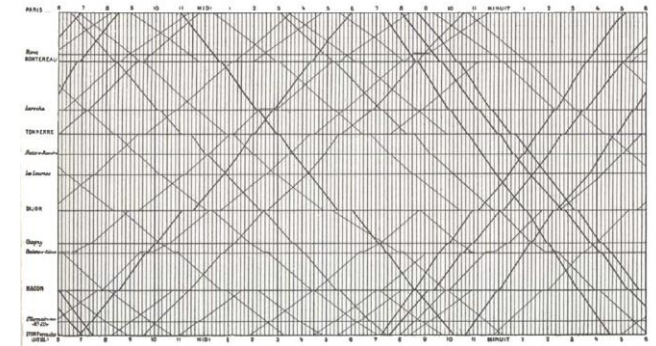
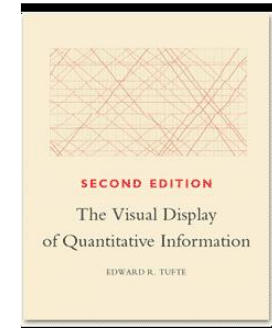
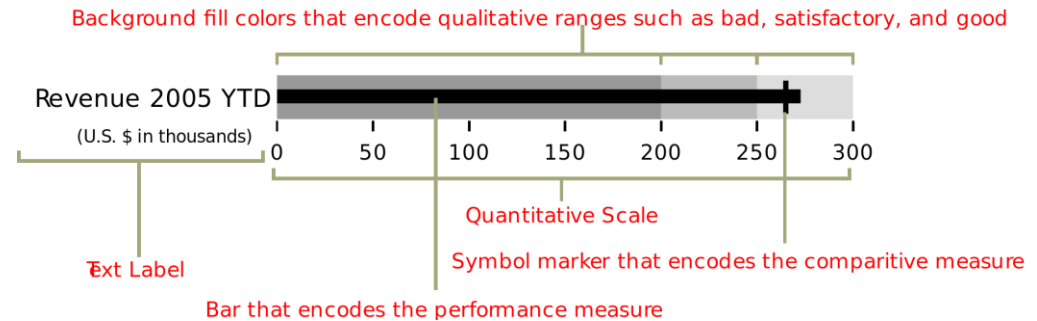
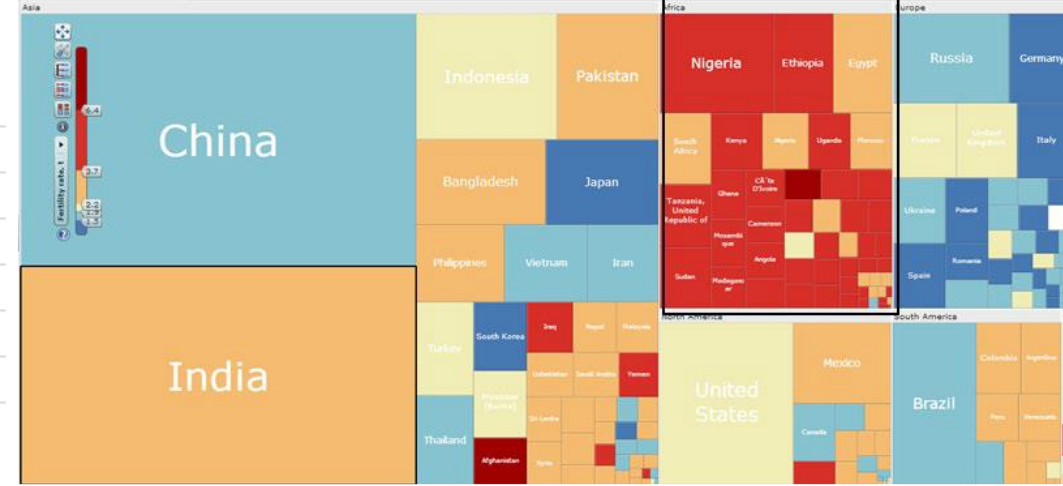
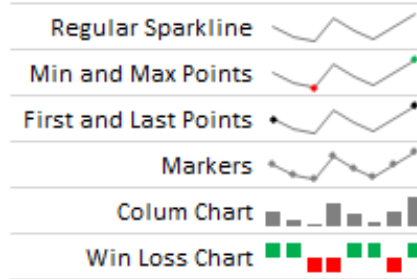


DIAGRAM TYPES

- Diagram types:
 - Bar charts
 - Scatter plots
 - Line charts
 - Sparklines
 - Box Plot
 - Spatial maps
 - Heatmaps
 - Treemaps
 - Bullet graphic

- Recommended reading:
 - Evergreen, S. D. H. (2019). Effective Data Visualization. London: Sage.
 - Kirk, A. (2019). Data Visualization. London: Sage.

Types of Sparklines



BEST PRACTICES

- **Organize** information to support its meaning and use.
 - Reading flow
 - Grouping
 - Gestalt laws
 - Enable or prevent comparisons
- Maintain **consistency** to enable quick and accurate interpretation
 - Top-down processing: use prior knowledge/experience and expectations
 - Differences generate attention
- Put **supplementary information** within reach
 - Outsource details
 - → interaction
- Expose **lower-level** conditions
 - For issues below the granularity level
 - Yes/No prompt is often sufficient, with the possibility for interaction
 - Exercise restraint!



BEST PRACTICES

- Make the experience **aesthetically** pleasing
 - Subtle color saturation
 - Few, deliberate contrasts
 - High resolution, readability
 - C.R.A.P.
- Prevent excessive **alerts**
 - Crying wolf
- Keep viewers **in the loop**
 - Maintain situational awareness
 - Don't rely solely on red/green traffic lights when comprehension is important
- Accomodate **real-time monitoring**
 - Update frequency; consider stopping if necessary
 - Consider audio alerts
 - Timestamp alerts



EXERCISE: SKETCH A DASHBOARD

Step 1: Select one of the following applications:

Corporate Performance Monitoring:

An information dashboard can be used to monitor the performance of a company in real-time. Various metrics such as revenue, expenses, profit margins, inventory levels, customer satisfaction, etc., can be visualized. This allows executives to quickly respond to trends, identify bottlenecks, and make strategic decisions.

Healthcare Analysis:

In healthcare, dashboards can be employed to track various aspects of patient care. This could include analyzing hospital occupancy, patient flows, treatment outcomes, medication availability, and more. Physicians and hospital administrators could use this information to make efficiency improvements and enhance the quality of care.

Traffic Monitoring and Management:

In urban areas, dashboards can be used to monitor and manage traffic. By integrating data from traffic cameras, sensors, and GPS systems, traffic flows can be analyzed, congestion can be detected, and alternative routes can be suggested. This enables traffic authorities to quickly respond to events, improve traffic efficiency, and enhance traffic safety.



EXERCISE: SKETCH A DASHBOARD

Step 2: Sketching your dashboard

Use sketching software or pen and paper to sketch your dashboard.

Think about the following questions:

- What level of detail should be offered? In case of doubt, it is better to offer too much detail than too little. Edward Tufte clearly recommends this: To clarify show detail!
- Do you consider the common mistakes in dashboard design mentioned by Stephen Few?

